

SELECT R1.A1, R2.A4
FROM R1, R2
WHERE R1.A2 = R2.A3

$R_1(A_1, A_2)$
 $R_2(A_3, A_4)$ SCHEMA

$\pi_{\langle A_1, A_4 \rangle} (\sigma_{A_2=A_3} (R_1 \times R_2))$

	A1	A2		A3	A4
R1	a	1	R2	1	h
	b	2		3	g
				2	h

risultato

A1	A4
a	h
b	h

$R(A, B, C, D)$ X (schema)

senza rimoz dupl.

con rimoz - dupl. (DISTINCT)

1	
2	SELECT dipartimento
3	FROM Impiegati I;

Data Output Messages Notific

Showing rows: 1 to 6

dipartimento
IT
Amministrazione
IT
IT
Marketing

Query	Query History
1	
2	SELECT DISTINCT dipartimento
3	FROM Impiegati I;

Data Output Messages Notifications

dipartimento
Marketing
Amministrazione
IT

	id	nome	cognome	mansione	dipartimento	stipendio
	[PK] integer	character varying (50)	character varying (50)	character varying (50)	character varying (50)	numeric (10,2)
1	1	Marco	Rossi	Analista	IT	35000.00
2	2	Laura	Bianchi	Sviluppatore	IT	32000.00
3	3	Giuseppe	Verdi	Dirigente	Amministrazione	60000.00
4	4	Anna	Ferrari	Analista	IT	36000.00
5	5	Paolo	Romano	Sviluppatore	IT	33000.00
6	6	Giovanna	Esposito	Dirigente	Marketing	58000.00

SELECT *
 FROM Impiegati;

```
(SELECT I.cognome
FROM Impiegati I
WHERE I.cognome LIKE '%si%')
UNION
(SELECT I.cognome
FROM Impiegati I
WHERE I.cognome LIKE 'R%');
```

```
(SELECT I.cognome
FROM Impiegati I
WHERE I.cognome LIKE '%si%')
INTERSECT
(SELECT I.cognome
FROM Impiegati I
WHERE I.cognome LIKE 'R%');
```

```
(SELECT I.cognome
FROM Impiegati I
WHERE I.cognome LIKE '%si%')
EXCEPT
(SELECT I.cognome
FROM Impiegati I
WHERE I.cognome LIKE 'R%');
```

Ordini

	id [PK] integer	clienteid integer	data date	importo numeric (10,2)
1	1	1	2025-01-15	150.00
2	2	1	2025-02-20	75.50
3	3	2	2025-02-10	320.75
4	4	4	2025-03-05	45.90
5	5	6	2025-03-12	200.00
6	6	3	2025-03-15	130.45

Clienti

	id [PK] integer	nome character varying (50)	cognome character varying (50)	email character varying (100)	telefono character varying (20)	città character varying (50)
1	1	Mario	Rossi	mario.rossi@email.it	345-1234567	Milano
2	2	Maria	Bianchi	maria.b@azienda.com	339-7654321	Roma
3	3	Antonio	Verdi	a.verdi@email.it	333-5555555	Napoli
4	4	Marco	Ferrari	marco2000@gmail.com	347-9876543	Milano
5	5	Martina	Romano	martina.r@outlook.it	348-1122334	Torino
6	6	Michele	Russo	michele@gmail.com	340-6677889	Roma

JOIN ORDINARIO

```
SELECT C.Nome, C.Cognome, O.ID AS NumeroOrdine, O.Data, O.Importo
FROM Clienti C
JOIN Ordini O ON C.ID = O.ClienteID
ORDER BY O.Data;
```

	nome character varying (50)	cognome character varying (50)	numeroordine integer	data date	importo numeric (10,2)
1	Mario	Rossi	1	2025-01-15	150.00
2	Maria	Bianchi	3	2025-02-10	320.75
3	Mario	Rossi	2	2025-02-20	75.50
4	Marco	Ferrari	4	2025-03-05	45.90
5	Michele	Russo	5	2025-03-12	200.00
6	Antonio	Verdi	6	2025-03-15	130.45

$\pi \left(\sigma_{\text{Ordini.clienteid} = \text{Clienti.id}} (\text{Clienti} \times \text{Ordini}) \right)$

LEFT JOIN

```
SELECT C.Nome, C.Cognome, O.ID AS NumeroOrdine, O.Data, O.Importo
FROM Clienti C
LEFT JOIN Ordini O ON C.ID = O.ClienteID
ORDER BY O.Data;
```

	nome character varying (50)	cognome character varying (50)	numeroordine integer	data date	importo numeric (10,2)
1	Mario	Rossi	1	2025-01-15	150.00
2	Maria	Bianchi	3	2025-02-10	320.75
3	Mario	Rossi	2	2025-02-20	75.50
4	Marco	Ferrari	4	2025-03-05	45.90
5	Michele	Russo	5	2025-03-12	200.00
6	Antonio	Verdi	6	2025-03-15	130.45
7	Martina	Romano	[null]	[null]	[null]